Technical Considerations in Dental Amalgam

INTENDED LEARNING OUTCOMES

By the end of lecture,the student should be able to

- **Explain** the manipulation technique of dental amalgam and restoration technique.
- **Describe** about mercury toxicity and how to prevent it in dental office

INTRODUCTION:

- Dental amalgam restorative material has been the primary direct restorative material for more than 100 years
- As of today, the popularity of amalgam as a direct restorative material has decreased.
- The decline is attributed in part to increased interest in tooth-colored composite resin restorations and the minimally invasive nature of their preparation steps.

SUCCESS OF RESTORATION

- ✓ **Proper follow of all the principles in cavity preparation**

and

- ✓ **Proper manipulation of amalgam restorative material**

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Success of restoration

TECHINICAL CONSIDERATIONS

For restoring a tooth with amalgam, the following

Steps for restoring with Amalgam

- Selection of alloy
- Proportioning of alloy and mercury
- Trituration
- Mulling
- Condensation
- Burnishing
- Carving
- Finishing & polishing.

1.ROBERT CRAIG & JOHN M.POWERS RESTORATIVE DENTAL MATERIALS
13TH EDITION pages 200-210

Selection of alloy

Factors Affecting the selection of alloy
✓**PARTICLE SIZE AND SHAPE**
✓**Low copper or High Copper (Elimination of gamma-2 phase)**
✓**PRESENCE (OR) ABSENCE OF ZINC**

- More than 90% amalgam currently used are HIGH COPPER ALLOYS

- ZINC CONTAING ALLOYS(More than 0.01% Zn)
Result in excess dimensional change during moisture contamination

ADVANTAGES OF HIGH COPPER AMALGAMS
NO GAMMA-2 PHASE
✓HIGH EARLY STRENGTH ✓LOW CREEP ✓GOOD CORROSION RESISTANCE ✓GOOD RESISTANCE TO MARGINAL FRACTURE

LATHE CUT ALLOYS	SPHERICAL ALLOYS
▪ ROUGH ,IRRREGULAR	▪ SMOOTHER

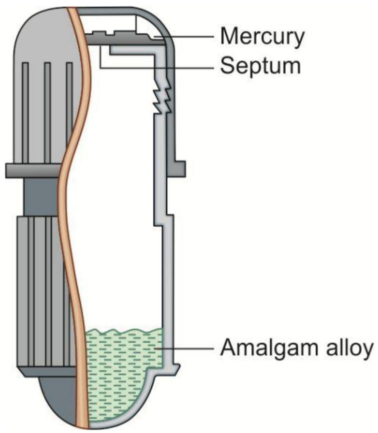
Proportioning of alloy to mercury

The amount of alloy and mercury to be used can be described as the **MERCURY/ALLOY RATIO** which signifies the parts by weight of mercury and of alloy to be used.

NOTE:

- Use of little mercury results in **dry mix**, impairs the strength of amalgam and reduces corrosion resistance.
- Use of excess mercury results in **increase in setting expansion**

❑Mercury/alloy ratio:



- Correct proportioning of alloy & mercury is essential for formulating a suitable mass of amalgam for placement into the prepared cavity.

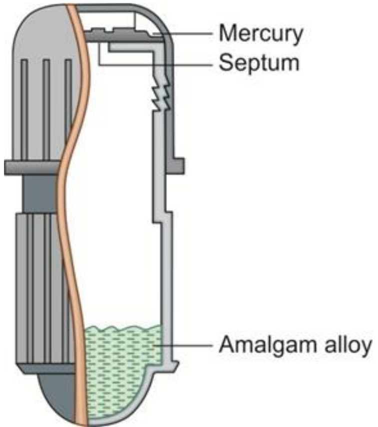
The recommended mercury/alloy ratio

- For lathe cut alloys is approximately **1:1 or 50% mercury**
- In case of spherical alloy **mercury should be 42%** *because spherical particles have lower surface/volume ratios* **(5:8)**

- AUTOMATED MECHANICAL DISPENSERS/**AMALGAMATOR**

- CURRENT RECOMMEDATION OF **"NO TOUCH PROCEDURE"**

- CAPSULES WITH **PRE-PROPORTIONED AMOUNTS** OF ALLOY AND MERCURY ARE USED.
- They are separated in the capsule by a membrane/septum
- Just before the mix is triturated(Mixed) the membrane is ruptured by compression of capsule prefitted with plastic pestle in rod or disc shape
- Capsules are color coded

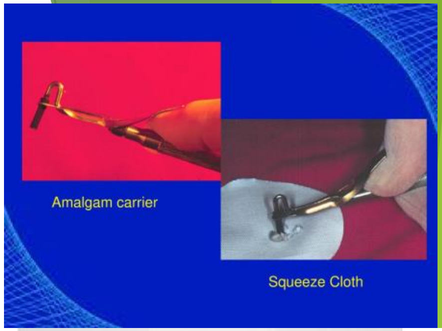


The following techniques were used for achieving to reduce mercury in final restoration

1. SQUEEZE CLOTH technique
2. INCREASING DRYNESS technique
3. EAMES technique

First technique- Squeeze cloth technique

Excess mercury is removed with the help of the cloth by SQUEEZING through it.



Second technique- Increasing dryness technique, during condensation of each increment a mercury rich soft layer comes to the surface. High condensation pressure will squeeze out the excess mercury.

- Condense amalgam thoroughly

Ideal condensation pressure is 3 to 4 lbs

The **most common method** to reduce mercury content is Minimal mercury technique or Eames technique(**MERCURY:ALLOY=1:1**) ,in which sufficient mercury will be present to form coherent and plastic mass after trituration.

Trituration

- The main objective of trituration is to provide proper amalgamation of the mercury and the alloy.

Types of mixing of amalgam

1. Hand mixing
2. Mechanical mixing

Originally the alloy & mercury were mixed or triturated by hand with mortar & pestle.

Today, however **mechanical amalgamators** are used which saves the time & standardized the procedure.

Hand mixing

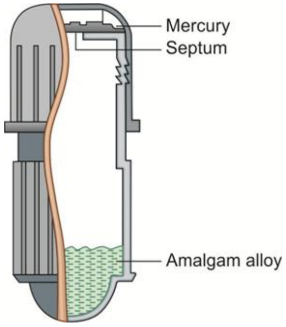
- A glass mortar and pestle are used.
- The mortar has its inner surface roughened to increase the friction .
- Usually a period of 25 to 45 seconds is sufficient for hand mixing.

MORTAR AND PESTLE

Mechanical trituration

- The alloy and mercury are present in the capsule,the capsule is secured in the amalgamator is turned on. There is an **automatic timer** for controlling the mixing time.

Modern amalgamator mixing time -7-8 secs



Amalgamator/ triturator

Note:

- Spherical alloy require less amalgamation time than lathe-cut alloys

- For a given alloy/mercury ratio, **increasing the trituration speed shortens the working and setting time**

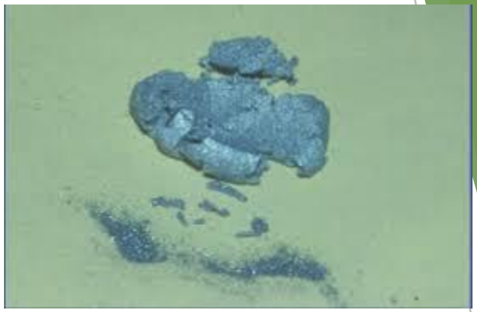
Consistency

- **Proper/normal mix** retains its luster longer after polishing and also remains warm when it is removed from the capsule.

MIXING VARIABLES

1.UNDER TRITURATED MIX

DULL,DRY,ROUGH AND GRAINY AND MAY CRUMBLE



- Tarnish and corrosion can occur
- Strength is less
- Mix hardens too rapidly
- Excess mercury still present

2.NORMAL MIX

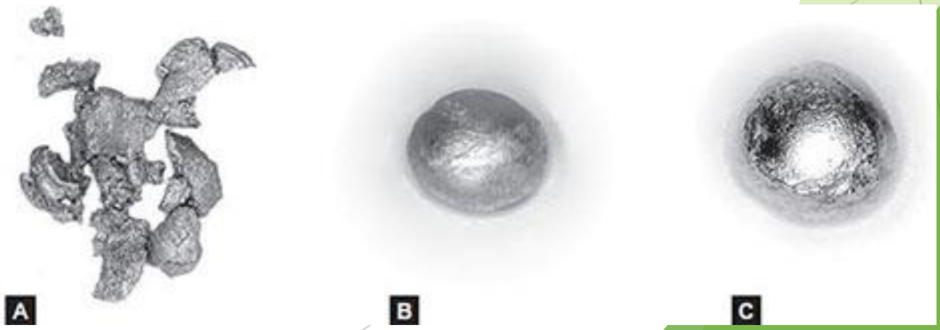
SHINY SURFACE,SOFT AND SMOOTH CONSISTENCY,SEPARATES IN SINGLE MASS FROM CAPSULE

- Warm when removed from capsule
- Best compressive and tensile strength
- Has luster after polishing
- Increased resistence to tarnish and corrosion

3.OVER TRITURATED MIX

SOUPY(WET), HOT SHINY MIX,DIFFICULT TO REMOVE FROM CAPSULE,STICKS TO CAPSULE

- Working time increased
- Higher contraction of amalgam
- Creep is increased



MULLING

- Mulling is actually a continuation of trituration.
- It increases the homogenity of the mass and get a single consistent uniform mix.
- The mix is enveloped in dry piece of rubber dam and rubbed between the first finger and thumb on a the palm of hand for 2-5 sec.

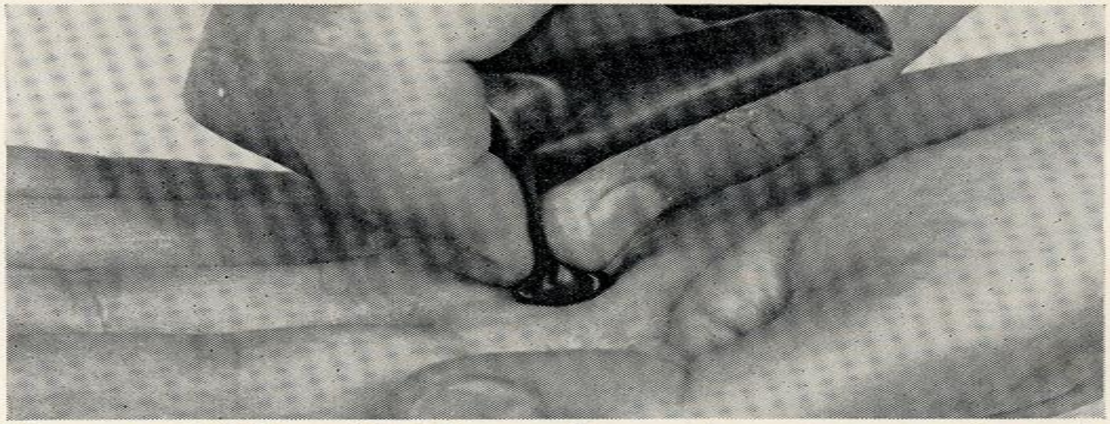
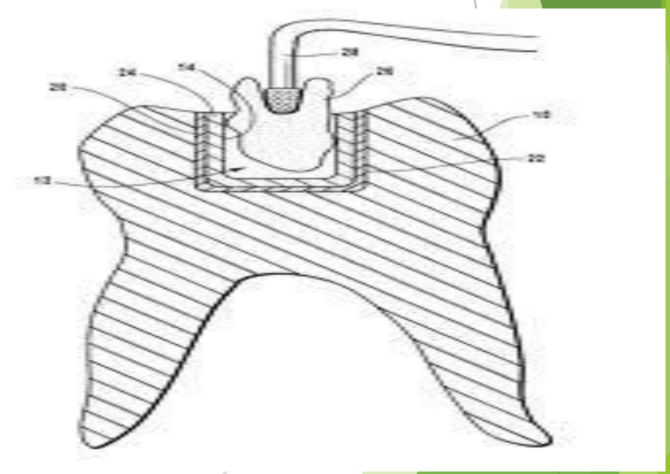


Figure 5-25. Mulling of amalgam mass in rubber dam.

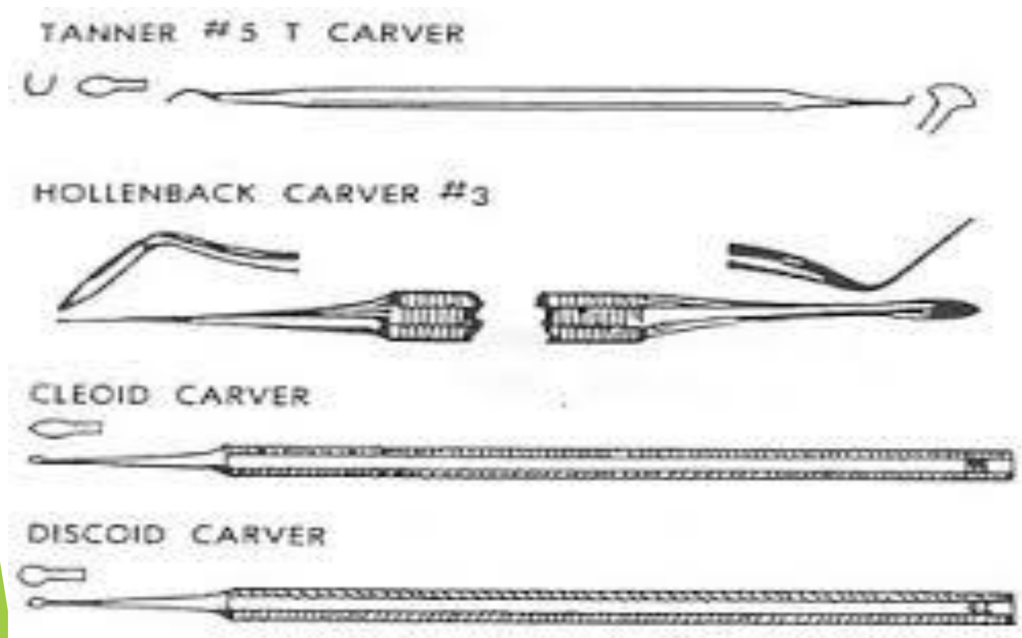
CONDENSATION

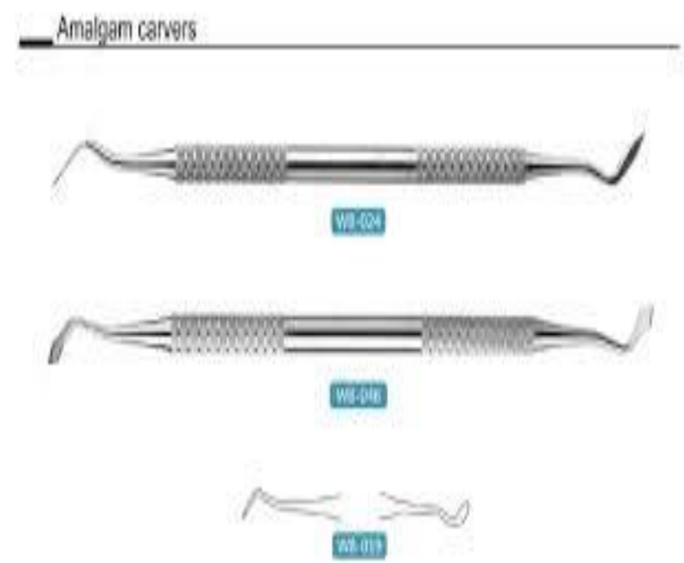
- The objective of condensation is to compact the alloy into the prepared cavity so that there is sufficient mercury present to ensure complete continuity of the matrix phase between the remaining alloy particles.(**coherent mass**)
- To reduce voids
- To minimise mercury
- To adapt correctly to all margins,walls, line angles of cavity



Armamentarium

- ❖ Amalgam Condensers
- ❖ Amalgam Burnishers
- ❖ Carvers
  - Cleoid or Discoid
  - Cone shaped
  - Diamond
  - Hollenback





- After the mix is made, condensation of the amalgam should be promptly initiated. *Condensation of partially set material probably fractures and break up the matrix that has already formed.*

- Condensation should be as rapid as possible and a fresh mix of amalgam should be made if condensation is intiated after 3-4 mins.

- The field of operation should be dry before condensation

EFFECT OF DELAY IN CONDENSATION

✓REDUCES PLASTICITY OF THE mix-cannot adapt well to cavity walls
✓RESULT IN AMALGAM RESTORATION WITH MORE MERCURY content-less compressive strength and more creep

EFFECT OF MOISTURE CONTAMINATION DURING CONDENSATION

DELAYED EXPANSION

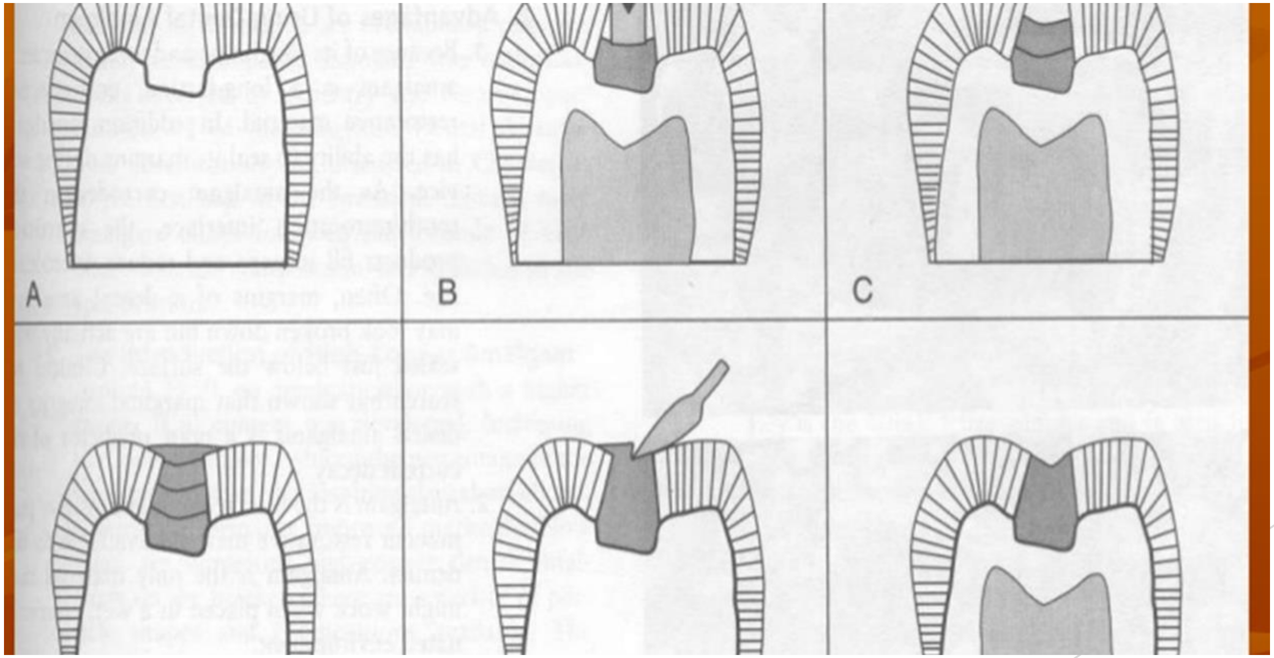
Hand condensation

- Once the increment of amalgam is inserted into the cavity preparation it should be condensed with pressure to avoid voids and to adapt the material to the walls, the condenser point is forced into the amalgam mass under hand pressure.
- Condensation is started at the center and then condenser point is **stepped** little by little towards the cavity wall.

Hand condenser

- After condensation of the each increment excess mercury should left over the first increment so that it can bond with the next increment.

- The procedure of adding an increment, condensing it, adding another increment and so forth is continued until the cavity is overfillled.(**STEPPING**)



Size of increment and force applied

- Size of the increments should be small,

In larger piece of increment, it is difficult to reduce the voids and to adopt the alloy to the cavity walls.

- Sufficient condensation force should be used to force the alloy particles together and to reduce voids

Condensation pressure

When a given force is applied, ***smaller the condenser greater the pressure exerted on the amalgam.***

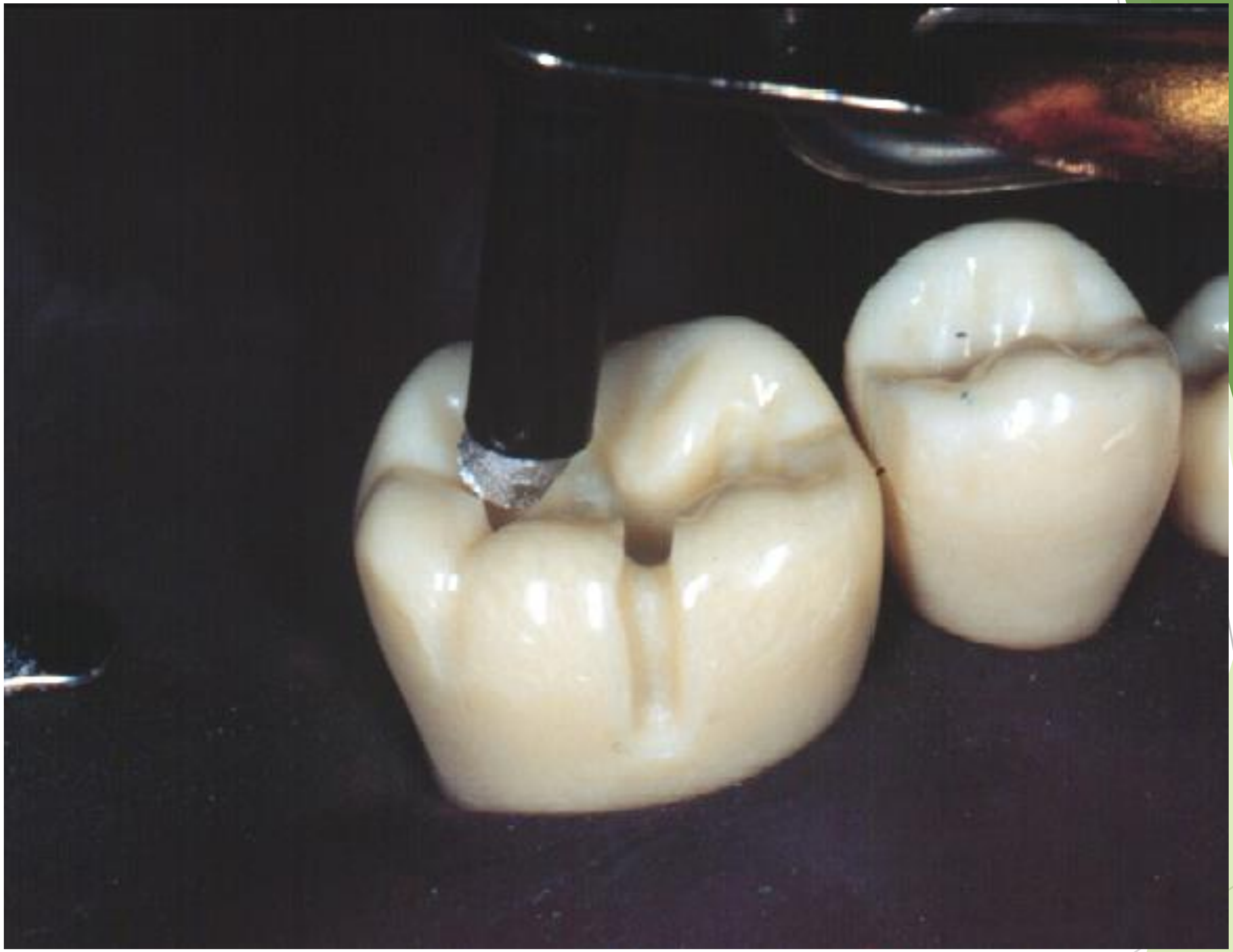
Ideal condensation pressure is 3 to 4 lbs **(1.4 - 1.8 kg)**

•Serrated condenser is preferred than round condenser in case of the corner of the cavity.

•The shape of condenser points should conform to the area under condensation.

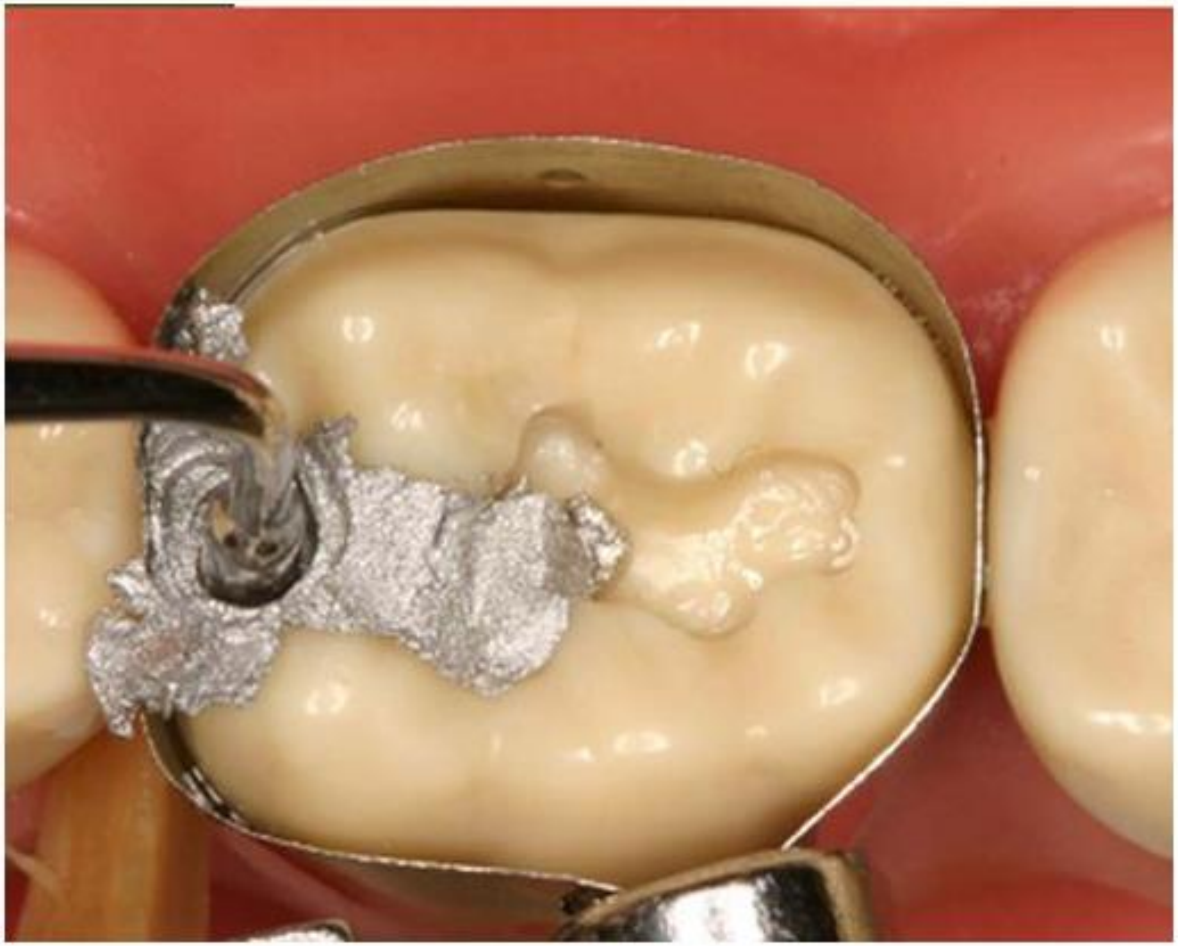
eg- a round condenser is ineffective in the corner of the cavity,a triangular or rectangular is indicated in such areas

Check condenser fit

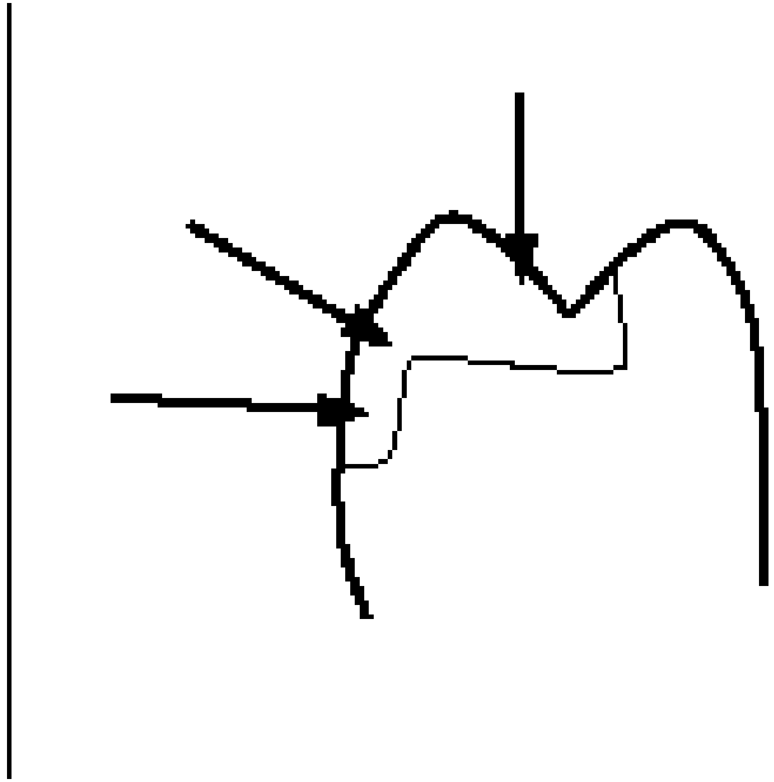


First increment

Lateral Condensation



- Condense amalgam thoroughly



Condense toward walls

Condense to Margins

Overpacked

Carving

- After amalgam is condensed in cavity it is carved to reproduce the proper tooth anatomy.
- **It should be started when the amalgam is hard enough to offer resistance to the carving**.(AMALGAM CRY)
- *Half side of the carver should rest on cuspal inclines and half side of carver should be on restoration*
- Intial carving consists of removal of excess bulk using cleoid carver

OBJECTIVES OF CARVING

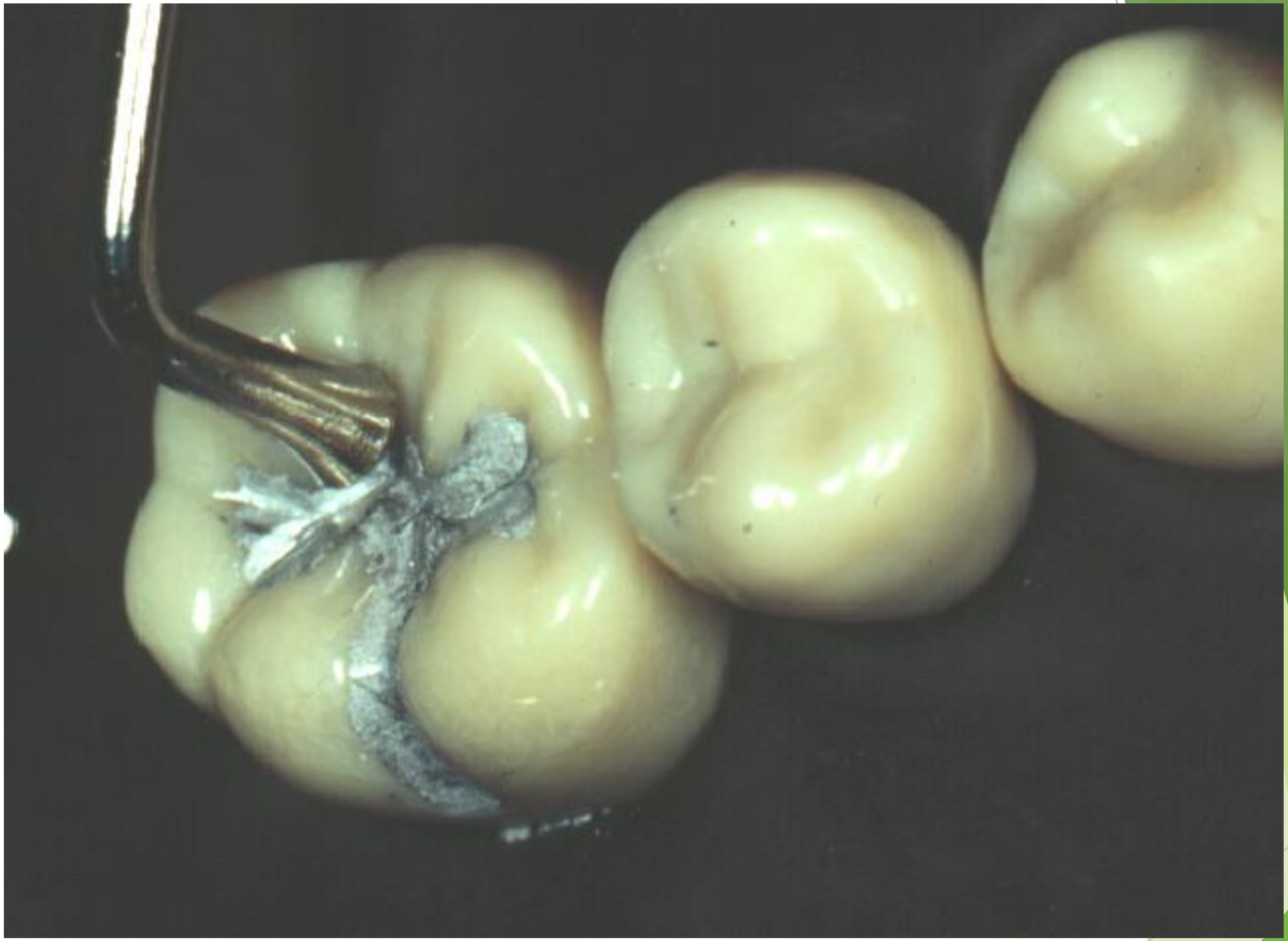
1. Functional non-interfering Occlusal anatomy

2. Adequate compatible marginal ridges, in class II restoration

3. Proper contours

4. MINIMAL flash(excess amalgam that results from overfilling)

Carve to margin



Final shape and burnishing

Completed restoration

Burnishing

Burnishing of the occlusal anatomy can be accomplished with the help of the **ball burnisher**.

<u>Objectives</u> of burnishing are

➢ Removes the scratches & irregularities on the amalgam surface

➢ Adapt material more to the margins

➢ Close the minute marginal discrepancies

➢ It is done by lightly rubbing the carved surface with a suitable size burnisher to improve the smoothness of the restoration.

- ✓**PRE-CARVE BURNISHING-**intiated just before carving
- ✓**POST CARVE BURNISHING**

Finishing and polishing

- Final smoothening can be done with the help of commercially available amalgam polishing system.

NOTE: During polishing, the restoration should be kept moist and only low rotational speeds with light intermittent pressure should be used, to avoid any over heating. Excessive heat production (more than 60^0C) during finishing and polishing can permanently damage the pulp

NOTE: The final finish of the restoration should not be done after 24 hours, set because crystallization is not complete and compressive strength is not completely attained

Polishing

MERCURY HAZARDS & MANAGEMENT:

- Mercury is toxic & proper precaution should be taken to limit exposure of the patient and the dental team to mercury or its vapor.
- Mercury hygiene is the process of handling mercury to minimize health risks.
- Hence good mercury hygiene will limit the exposure from all these routes.

Mercury exposure in dental office results from :

- Incorrect storage of waste amalgam.
- Spillage of mercury.
- Mixed but unhardened amalgam during trituration, insertion, & setting.
- During finishing & polishing.
- During removal of amalgam.

Mercury Allergy:

- Allergy to mercury is very rare.
- Occupational contact dermatitis can occur due to contact with mercury; this is avoided by using latex gloves & no touch technique.

PREVENTION OF MERCURY TOXICITY

DENTAL MERCURY HYGIENE RECOMMENDATIONS:

1.Provide proper ventilation in the dental office by having fresh air exchanges and periodic replacement of air conditioning filters which act as traps for mercury.

2.Monitor the Hg vapor levels in the office and personnel periodically using **dosimeter badges** and vapour analyzer (permissible limit -50µg/m^3)

3. Use proper work area or dental office design to facilitate **spill containment and cleanup.**

4.Use pre-capsulated alloys to prevent the possibility of bulk mercurcy spillage in office during manipulation

5.Proper Hg:alloy ratio to avoid the need to remove excess mercury before condensation.

6.Use of **amalgamators** fitted with cover to prevent spillage.

7. **Use care in handling amalgam** and avoid direct skin contact with mercury or freshly mixed amalgam as it may cause skin allergy.

8.Use of high volume evacuation when finishing or removing amalgam. Evacuation systems have traps or filters which should be checked, cleaned and replaced periodically.

9.Change facemasks as necessary when removing old amalgam restoration in patient.

10.Dispose of mercury contaminated items **in sealed bags** according to applicable regulations.

11.Store amalgam scrap/waste (unused amalgam remaining after a procedure)in tightly closed container either dry or under radiographic fixer solution.

12. Clean up spilled mercury properly **using trap bottles, tape** or freshly mixed amalgam to pick up droplets, and commercial spill cleanup kits. Do not use a household vacuum cleaner.

13.Avoid carpet floor coverings in dental office as it traps mercury and amalgam waste.

14.Use of rubber dam to prevent mercury contamination during manipulation of amalgam and polishing.Always avoid dry polishing of amalgam as it releases mercury vapours into the office.

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15.Always follow strict infection control and personnel barrier protection with facemasks,gloves and eyewear.

16.Remove all professional clothing before leaving workplace

Thank you…….

LEARNING RESOURCE

1.Robert craig & john m.Powers Restorative dental materials

13th edition

Page no:-**200-210**

2.Sturdevants "Art & Science of operative dentistry"

6th edition

Page no:- **14 – 31,403-406**